

Financial Time Series

TA Session #6

April 30, 2021

$(A-I)X_{t-1}$ is stationary

BX_{t-1} is stationary

$(b_{ij}) \begin{pmatrix} \cdot & X_{1,t-1} \\ & \vdots \\ & X_{m,t-1} \end{pmatrix}$ is stationary

$\text{Rank}(B) = R$ B is $m \times m$

$R \begin{Bmatrix} \vdots \\ \vdots \\ \vdots \\ 0 \end{Bmatrix} \begin{pmatrix} X_{1,t-1} \\ \vdots \\ X_{m,t-1} \end{pmatrix}$ is stationary

R linearly indep
cointegrating vectors

①

Linear combinations of weakly stationary processes are weakly stationary

$$\underbrace{\alpha_1 X_{1t}}_{\text{weakly stat}} + \underbrace{\alpha_2 X_{2t}}_{\text{weakly stat}}$$

$$\alpha_i = (\alpha_{i1}, \dots, \alpha_{im})$$